3	(a)		Coherence means that the phase difference between two waves remain constant and do not vary with time.	
	(b)			
		(i)	The microwaves waves from the two sources undergo interference. The path/phase difference at P is changing as S_1 moves. When the waves are in phase at P such that the path difference between the sources is an integer number of wavelengths (or equivalent statement about phase difference); constructive interference occurs at P resulting in maximum intensity. When the waves are anti-phase at P when the path difference is an integer number of wavelengths plus half a wavelength (or equivalent statement about phase difference), destructive interference occurs at P resulting in minimum/zero intensity.	
		(ii)	path difference = XY = $\frac{\lambda}{2}$ λ = 0.164 m $v = f\lambda$ $f = \frac{3.0\times10^8}{0.164} = 1.8 \times 10^9\ Hz$	
		(iii)	As the microwaves from S1 and S2 are no longer have the same axis of polarization, destructive interference does not take place (or the waves do not cancel out). Hence, the intensity of the microwave at P is now between the minimum and the maximum.	
	(c)			
		(i)	Original amplitude of S_1 = 12 units As intensity is proportional to the square of the amplitude New amplitude of S_1 = $\sqrt{\frac{I/2}{I}} \times 12$ = 8.5 units.	
		(ii)	Wave in antiphase with original but with only 8.5 units	